ASSIGNMENT - 2

Cloud Computing

(Attempt any 50 questions)

Section 1: Cloud Computing Fundamentals

1. What is cloud computing? Explain its main characteristics.
2. Describe on-demand self-service and explain why it is useful.
3. Explain broad network access in the context of a privately managed cloud environment.
4. What is resource pooling? Give an example from a university data centre.
5. Explain rapid elasticity and how it differs from manually purchasing new hardware.
6. Define measured service. Identify three resources that an organization might measure.
7. Differentiate cloud computing from traditional standalone computing.
8. Explain the difference between scalability and elasticity.
9. Describe two situations in which an organization might choose a privately managed cloud.
10. Write a short note on the evolution from physical servers to virtualized infrastructure.

Section 2: Cloud Architecture and Service Models

11. Draw and explain a basic cloud computing architecture showing clients, network, management layer, and compute/storage resources.
12. Explain the roles of the front end and back end in a cloud system.
13. What is a cloud management plane? List four functions it may perform.
14. Explain Infrastructure as a Service (IaaS) using an organization-owned private cloud example.
15. Explain Platform as a Service (PaaS) and identify the tasks it can simplify for developers.
16. Explain Software as a Service (SaaS) using an application hosted within an organization's own infrastructure.
17. Differentiate IaaS, PaaS, and SaaS by describing what the provider/operator and user manage.
18. What is a service-level agreement (SLA)? List five items that may be included in one.
19. Explain multi-tenancy and describe how tenant isolation can be maintained inside a private cloud.
20. Describe the roles of a cloud administrator, cloud architect, and cloud user.

Section 3: Deployment and Private Cloud

21. Define a private cloud. Explain how it differs from a conventional server room.
22. List the essential components required to build a private cloud in an organization.
23. Explain the difference between on-premises infrastructure and a private cloud.
24. Describe two advantages and two limitations of a private cloud.
25. What is a community cloud? Give an example involving organizations with shared requirements, without using a public-cloud service.
26. Explain why governance and ownership are important when designing a private cloud.
27. Describe how a university could use a private cloud to provide virtual laboratories to students.
28. Identify five factors to assess before an organization builds a private cloud.

29. Explain capacity planning for a private cloud and identify the metrics that should be monitored.
30. Case study: A hospital wants to host sensitive records on infrastructure it controls. Suggest a high-level private-cloud design and justify your choices.

Section 4: Virtualization and Compute

31. Define virtualization and explain how it supports cloud computing.
32. Differentiate a Type 1 hypervisor from a Type 2 hypervisor.
33. What is a virtual machine (VM)? Identify its main components.
34. Explain the difference between a host operating system and a guest operating system.
35. Compare virtual machines and physical servers in terms of isolation, utilization, and management.
36. What are CPU overcommitment and memory overcommitment? Describe one risk of each.
37. Explain live migration of a virtual machine and list two prerequisites.
38. What is a VM snapshot? Explain why a snapshot is not a complete substitute for a backup.
39. Describe resource allocation for a VM running a database application.
40. Scenario: Several VMs are slow because they share one host. Explain how you would investigate and resolve the problem.

Section 5: Storage and Data Management

41. Explain the difference between block, file, and object storage.
42. Compare direct-attached storage (DAS), network-attached storage (NAS), and storage area networks (SANs).
43. What is thin provisioning? Explain one benefit and one risk.
44. Define storage redundancy and explain why it matters.
45. Differentiate backup, replication, and archiving.
46. Explain the 3-2-1 backup rule and how an organization could apply it within its controlled infrastructure.
47. What are Recovery Point Objective (RPO) and Recovery Time Objective (RTO)?
48. Describe a suitable storage design for a departmental file-sharing system.
49. Explain data deduplication and compression. How are they different?
50. Scenario: A private cloud's storage is nearly full. Propose a sequence of steps to investigate and manage capacity safely.

Section 6: Cloud Networking

51. Explain the role of networking in a cloud environment.
52. Define a virtual network and explain how it differs from a physical network.
53. What are virtual switches and virtual routers?
54. Explain VLANs and how they can help separate workloads.
55. Differentiate a subnet, IP address, gateway, and DNS server.
56. Explain software-defined networking (SDN) at a conceptual level.
57. What is a firewall? Describe how firewall rules can limit communication between cloud workloads.
58. Explain load balancing and give an example of when it is useful in a private cloud.
59. Describe the purpose of VPNs when administrators need secure remote access to an organization's infrastructure.
60. Scenario: A VM can reach other machines by IP address but cannot reach them by hostname. Identify likely causes and troubleshooting steps.

Section 7: Containers and Application Deployment

61. What is a container, and how does it differ from a virtual machine?
62. Explain the role of a container image and a container runtime.
63. What is a container registry? Why should access to it be controlled?
64. Define orchestration in the context of containers.
65. Explain the purpose of a pod, service, and deployment in a container orchestration system.
66. Compare deploying an application manually on VMs with deploying it using containers.
67. What is a microservices architecture? State two potential benefits and two challenges.
68. Explain configuration management and why configuration should be separated from application code.
69. Describe a basic deployment pipeline from source code to testing and release within an organization's own infrastructure.
70. Scenario: A containerized application works on a developer's laptop but fails in the data centre. List likely causes and a systematic debugging approach.

Section 8: Security, Privacy, and Governance

71. Identify the main security objectives of confidentiality, integrity, and availability, with one cloud-related example for each.
72. Explain identity and access management (IAM) and the principle of least privilege.
73. Differentiate authentication and authorization.
74. What is multi-factor authentication (MFA), and why is it useful for cloud administrators?
75. Explain encryption at rest and encryption in transit.
76. What is network segmentation? Explain how it can reduce the impact of a security incident.
77. Describe the purpose of audit logs and list four events that should be logged.
78. Explain the shared-responsibility concept in a privately operated cloud, distinguishing infrastructure-team duties from application-owner duties.
79. List five security controls for protecting virtual machine images and templates.
80. Scenario: An employee account has been compromised. Outline immediate containment, investigation, recovery, and prevention steps.

Section 9: Reliability, Monitoring, and Operations

81. Define availability and explain how it can be measured.
82. What is fault tolerance? Give an example involving a private-cloud service.
83. Differentiate high availability and disaster recovery.
84. Explain the purpose of monitoring CPU, memory, storage, and network utilization.
85. Differentiate metrics, logs, and traces.
86. What is alert fatigue? Suggest three ways to reduce unnecessary alerts.
87. Explain patch management and why maintenance windows may be required.
88. Describe a change-management process for updating a critical cloud platform.
89. Explain capacity forecasting and identify two risks of inaccurate forecasts.
90. Scenario: A service becomes unavailable after a planned update. Describe the troubleshooting, rollback, communication, and post-incident review steps.